

Effects of blood meal source and seasonality on reproductive traits of *Culex quinquefasciatus* (Diptera: Culicidae)

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

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Reviewed preprint posted

September 15, 2023 (this version)

Posted to bioRxiv

July 2, 2023

Sent for peer review

June 14, 2023

Kevin Alen Rucci , Mariana Pueta, Adrián Díaz 

Laboratorio de Arbovirus, Instituto de Virología “Dr. J. M. Vanella” (InViV), Facultad de Ciencias Médicas (FCM), Universidad Nacional de Córdoba (UNC) • Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina • Instituto de Investigaciones en Biodiversidad y Medioambiente (INIBIOMA), Universidad Nacional de Comahue (UnCo)

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Abstract

Host selection by mosquitoes is a keystone to understand viral circulation and predict future infection outbreaks. *Culex* mosquitoes frequently feed on birds during spring and early summer, shifting into mammals towards late summer and autumn. This host switch may be produced by changes in mosquito fitness. The aim of this study was to assess if blood meal source and seasonality may influence reproductive traits of *Culex quinquefasciatus* mosquitoes. For this purpose, *Cx. quinquefasciatus* mosquitoes were reared in simulated summer and autumn conditions and fed with two different hosts, chickens and mice, in a factorial design. Fecundity, fertility and hatchability during two consecutive gonotrophic cycles were estimated. We found a greater fecundity and fertility for mosquitoes fed upon birds than mammals. Fecundity and fertility increased in autumn for chicken-fed mosquitoes, whereas they decreased for mouse-fed mosquitoes. These traits decreased in the second gonotrophic cycle for mouse-fed mosquitoes, whereas they did not vary between cycles for chicken-fed mosquitoes. Blood meal source had a significant effect on hatchability, finding the lowest hatchability in mouse-fed mosquitoes during the second gonotrophic cycle, whereas no differences were detected among the remaining treatments. These results indicate that fecundity and fertility are influenced by blood meal source and seasonality, increasing towards autumn for mosquitoes fed on birds and decreasing for mosquitoes fed on mammals. Hatchability seems also being influenced by blood meal source. Our results suggest that blood meal source and seasonality induce metabolic changes in the mosquito reproductive physiology which might affect host selection patterns.

eLife assessment

This **useful** study presents **convincing** evidence that blood meal source and season affect *Culex quinquefasciatus* mosquito reproduction. Its unique focus on the interactive effects of both factors on mosquito fitness is of considerable relevance for the field, but the work suffers from **inadequate** experimental design - no replication, a population mismatch with the hypothesis region, and small sample sizes, limitations that were not sufficiently acknowledged in the discussion. The work will be of interest to those studying malaria and vector-borne diseases.

Introduction

The southern house mosquito, *Culex quinquefasciatus* Say (Diptera: Culicidae), is a worldwide distributed vector of several pathogens, such as *Wuchereria bancrofti* Cobbold (Kasili et al., 2009 [↗](#)), *Dirofilaria immitis* Leidy (Labarthe et al., 1998 [↗](#)) and avian malaria (Farajollahi et al., 2011 [↗](#)). Besides, this mosquito is also one of the main responsible for the transmission of the arboviruses West Nile virus (WNV) and Saint Louis encephalitis virus (SLEV), among others (Farajollahi et al., 2011 [↗](#)). Because of this, this species is of critical concern due to its impact on both public and veterinary health.

Since arboviruses are transmitted by the bite of infectious mosquitoes between vertebrate hosts, activity of these pathogens is determined, at least in part, by the blood feeding habits of these vectors. Host selection of *Culex* mosquitoes is variable and it can be classified in four categories: (1) mammophilic (Muturi et al., 2008 [↗](#)), (2) ornithophilic (Jansen et al., 2009 [↗](#)), (3) herpetophilic (Janssen et al., 2015 [↗](#)) and (4) generalist (Molaei et al., 2007 [↗](#)). Moreover, some species may experience a host switch in its feeding habits, from avian to mammal hosts (Hancock & Camp, 2022 [↗](#)). These species are of greater concern from an epidemiological standpoint because they can act as bridge vectors in the transmission of some arboviruses, such as SLEV and WNV (Kilpatrick et al., 2005 [↗](#)).

Culex quinquefasciatus is considered an ornithophilic species in several areas around the globe (Takken & Verhulst, 2013 [↗](#)). Despite its ornithophilic nature, *Cx. quinquefasciatus* and other *Culex* species could feed primarily from mammal hosts, including the human, in certain situations. Different studies have pointed out that in autumn there is an increased activity of SLEV and WNV in human populations (Kilpatrick et al., 2006 [↗](#), Spinsanti et al., 2008 [↗](#)), suggesting that viral activity spills over to mammals from birds as a result of the mosquito feeding host switch (Edman & Taylor, 1968 [↗](#), Kilpatrick et al., 2006 [↗](#)). A few hypotheses have attempted to explain why primarily ornithophilic mosquitoes also feed upon mammals in some contexts. One of the most widely accepted hypothesis states that the shift on feeding behavior of *Culex* mosquitoes is a consequence of the autumn migration of American robin (*Turdus migratorius*), the principal host in the northeastern US, generating an opportunistic change into mammal hosts (Kilpatrick et al., 2006 [↗](#)). However, because this avian species is lacking in Southern latitudes it is an implausible hypothesis for many austral areas, being restricted only for some regions of the US. A second hypothesis proposed by Burkett-Cadena et al. (2011) [↗](#) ensures that host breeding cycles drive the host shift of mosquitoes, and peaks of host use are detected in periods of reproductive investment. According to this hypothesis, in hosts with parental cares, during reproductive seasons there is a greater investment of energy in assuring the offspring survivorship, consequently producing an increase in susceptibility of being bitten by mosquitoes as a result of a decrease in defensive behaviors (Burkett-Cadena et al., 2011 [↗](#)). Other hypotheses suggest that mosquito host shift is a result of changes in host preference throughout the seasons (Edman, 1974 [↗](#)), changes in density

and feeding success of mosquitoes (Nelson et al., 1976 [↗](#)) or seasonal changes in host habitat use by mosquitoes, however, they were poorly or not supported by field data (Edman, 1974 [↗](#); Burkett-Cadena et al., 2011 [↗](#)).

Several biological (stress, metabolic rate, blood meal source) and environmental (temperature, photoperiod, i.e., season) variables that influence mosquito physiology might have an effect on reproduction traits, such as fecundity, development rate and survivorship, generating new nutritional requirements (Ciota et al., 2014 [↗](#), Costanzo et al., 2015 [↗](#), Gervasi et al., 2016 [↗](#); Yan et al. 2017 [↗](#), 2018 [↗](#)). These factors, ultimately, could produce seasonal variation in host use by mosquitoes. The aim of the present study is to evaluate the effect of both host blood meal source and seasonality (combination of temperature and photoperiod) on three reproductive traits (fecundity, fertility and hatchability) of *Culex quinquefasciatus* mosquitoes. We hypothesize that the seasonal host switch of mosquitoes is the result of the effect of blood meal source and seasonality on reproductive outputs. We expect a greater number of eggs (fecundity) and larvae (fertility) in mosquitoes fed on mammal in autumn than in an avian host, expecting the opposite pattern in summer.

Materials and methods

Establishment and rearing of mosquito colony

Egg rafts of *Culex quinquefasciatus* were collected from a drainage ditch at Universidad Nacional de Córdoba Campus, Córdoba city in February 2021. Each raft was maintained individually in plastic containers with one liter of distilled water. The hatched larvae were fed with 100 mg of liver power three times per week until pupation. Pupae were transferred to a plastic emerging cage (21 cm x 12 cm) covered with a tulle-like fabric, containing distilled water and without food. Adults emerging from each raft were identified morphologically (Darsie, 1985 [↗](#)) and molecularly (Smith & Fonseca, 2004 [↗](#)) to make sure they corresponded to *Cx. quinquefasciatus* since in Córdoba is also present *Cx. pipiens* and its hybrids (Branda et al., 2021 [↗](#)). All adults were reared in a cardboard cage of 22.5 liters (28 cm x 36.5 cm) and fed *ad libitum* with a 10% of sugar solution soaked in cotton pads on plastic glasses. For long-term maintenance of the colony, 24-h starved mosquitoes were offered with a blood meal of a restrained chicken twice a month. Four days post-feeding, a plastic recipient with distilled water was located inside the cage to allow engorged females lay egg rafts. Batches were collected and transferred to plastic containers (30 cm x 25 cm x 7 cm) in a proportion of 3 rafts per container, filled with 3 liters of distilled water. The hatched larvae were also fed with liver power at the same proportion described above. Pupae were transferred to emerging cages and adults to the final cardboard cage. The colony was inbred and maintained during several generations (generation > F₂₀) in the Insectary of the Instituto de Virología “Dr. J.M. Vanella” (InViV). Room controlled conditions were 28°C, 12L:12D photoperiod and 70% relative humidity.

Experimental design

Blood source

For experimental trials, an avian and mammal model was used to evaluate the effect of blood meal source on fecundity, fertility and hatchability. The former host was represented by live chicks from the species *Gallus gallus* and the latter by live mice of the species *Mus musculus* (strain C57BL/6). For each trial, 24-h starved adult female mosquitoes were provided with a blood meal from restrained live chicks or mice. Chicks were donated by the Bartolucci poultry farm (Córdoba, Argentina). Mice were commercially obtained at the Instituto de Investigación Médica Mercedes y Martín Ferreyra (CONICET - Universidad Nacional de Córdoba). Vertebrate hosts were offered 1 hour before the lights went out and kept for 3 hours during 2 consecutive gonotrophic cycles.

The experimental use of animals (*Gallus gallus* and *Mus musculus*) was approved by the ethical committee at Facultad de Ciencias Médicas, Universidad Nacional de Córdoba (FCM-UNC) in compliance with the legislation regarding the use of animals for experimental and other scientific purposes (accession code: CE-2022-00518476-UNC-SCT#FCM).

Seasonality

To assess the effect of seasonality (photoperiod + temperature) on fecundity, fertility and hatchability, a “typical” summer and autumn day from Córdoba city were simulated in an incubator where mosquitoes were kept throughout the entire experiment. Summer conditions were: $T^{\circ}_{\min} = 22^{\circ}\text{C}$, $T^{\circ}_{\max} = 28^{\circ}\text{C}$, photoperiod = 14L:10D. Autumn conditions were: $T^{\circ}_{\min} = 16^{\circ}\text{C}$, $T^{\circ}_{\max} = 22^{\circ}\text{C}$, photoperiod = 10L: 14D. Humidity: 60-70% for both conditions. Data were obtained from Climate Data website (<https://es.climate-data.org/>)

Feeding trial design

Four experimental colonies were established from egg rafts collected from the maintenance colony, based on the combination of blood source and seasonality: bird-summer, bird-autumn, mammal-summer and mammal-autumn. These experimental colonies were reared at controlled room conditions from eggs to adults to ensure all adults have the same size (Kauffman et al., 2017), then they were transferred to the incubator to begin the assay. Adults were kept in cardboard cages of 8.3 liters (21 cm x 24 cm), fed *ad libitum* with a 10% of sugar solution soaked in cotton pads on plastic glasses.

Experiments were performed during two consecutive blood meals at 5 and 14 days post-emerging. After each blood meal, female mosquitoes were anesthetized with CO_2 and classified as full engorged (VI Sella's stage), partially fed (I-V Sella's stage) and unfed (I Sella's stage) (Silva Santos et al., 2019). Full engorged females were counted and separated into another cage to complete its gonotrophic cycle. Four days post-feeding a glass with distilled water was located inside the cage to allow oviposition. After each oviposition cycle egg rafts were collected, counted, and transferred to a 12-well plate with 3 mL of distilled water, where they were photographed to posteriorly count the number of eggs per raft. Rafts were maintained in each well until they hatched into L1 larvae, time where they were fixed with 1 mL of 96% ethanol. After hatching, the number of L1 larvae per raft was counted.

Statistical analysis

The reproductive outputs measured for each experimental colony and gonotrophic cycle were fecundity, fertility and hatchability. All analyses were performed using R Studio statistic software, v.4.2.1 (R Core Team, 2022). Fecundity was defined as the number of eggs per raft and fertility as the number of L1 larvae hatched per raft. To evaluate the effect of blood meal source and seasonality on these two variables, a Generalized Linear Model (GLM; package MASS) with negative binomial error distribution and logarithmic link function was adjusted (Venables & Ripley, 2002), with fecundity and fertility as response variables, respectively. Blood source, seasonality and gonotrophic cycle in a three-way interaction were set as explanatory variables. Hatchability (= hatching rate) was defined as the ratio of the number of larvae to the number of eggs per raft. Because data did not follow any known exponential family probability distribution (needed for linear regression) a null model analysis via data randomization was performed. Prior to this analysis we adjusted a factorial ANOVA using hatchability as response variable and blood source, seasonality and gonotrophic cycle in a three-way interaction as explanatory variables. From this model we obtained and saved the observed F value (F_O) for each variable (main effects) and its combination (interaction effects). Posteriorly, we randomly permuted the response variable 10000 times and obtained simulated F value to create an own distribution for the F

parameter. The F_0 value was compared to the simulated distribution to test the null hypothesis that data were generated randomly against the alternative that there might be a pattern generated for at least one of the explanatory variables.

Results

Effect of blood source and seasonality on fecundity and fertility

Blood meal source had a significant effect on fecundity but not in fertility. Seasonality had a significant effect on both fecundity and fertility. Gonotrophic cycle did not show an effect on neither of the two response variables (Tables 2, 3).

It was observed an interaction effect between blood meal source and seasonality on fecundity and fertility (Table 2, 3). For both response variables, it was not found differences in the number of eggs and larvae per raft in summer for both blood meal sources, chicken and mouse. However, in autumn it was detected an increase in fecundity and fertility for bird-fed mosquitoes and a decrease for mouse-fed mosquitoes (Fig. 1 A, C, Table 1). In general, avian-fed mosquitoes presented greater fecundity and fertility than mammal-fed mosquitoes, regardless of the season. However, from summer to autumn, it was observed an increase of 13% and 18% on fecundity and fertility, respectively, for chicken-fed mosquitoes and a decrease of 17% and 19%, respectively, for mouse-fed mosquitoes.

It was also detected an interaction effect between blood meal source and gonotrophic cycle (Table 2, 3). It was observed a decrease of 29% in fecundity and fertility from the first to second cycle for mouse-fed mosquitoes, being this treatment different from the remaining groups, in terms of fitness (Fig. 1 B, D, Table 1). During the second gonotrophic cycle fecundity and fertility for chicken-fed mosquitoes was 41% and 40% greater, respectively, than mosquitoes fed upon mice. It was not found statistically significant differences among the other three groups.

The rest of single factors and interactions from the fecundity and fertility models were not statistically significant (Table 2, 3).

Effect of blood source and seasonality on hatchability

By visual inspection it was detected that mosquitoes fed upon mammals possessed the greater hatchability, except for mosquitoes feeding on mammals in autumn during the second gonotrophic cycle, which registered the least hatching rate (Fig. 2). Statistically, it was detected a significant effect of blood meal source on hatching rate ($F_0 = 0.00074$, $p = 0.0004$). Highest hatchability values were observed in mouse-fed mosquitoes, ranging from 82% (autumn, second gonotrophic cycle) to 96% (autumn, first gonotrophic cycle) (Table 1, Fig. 2).

Discussion

Several host physiological factors have been studied as putative candidates to explain the changes in mosquito feeding behavior, including stress, health status or metabolic rate (Gervasi et al., 2016; Yan et al. 2017, 2018). However, it has not been investigated how seasonal use of hosts affects reproductive traits in mosquitoes. It is known that some factors like host blood source, temperature or photoperiod have an effect on some reproductive aspects (Takken & Verhulst, 2013). Some authors have studied and proved the significant effects of blood meal source (Telang & Skinner, 2019), temperature (Ciota et al., 2014) and photoperiod (Costanzo et al., 2015) on fecundity and fertility. These studies illustrate how single effects of blood source and environmental factors may influence reproductive outputs. However, to our knowledge, there

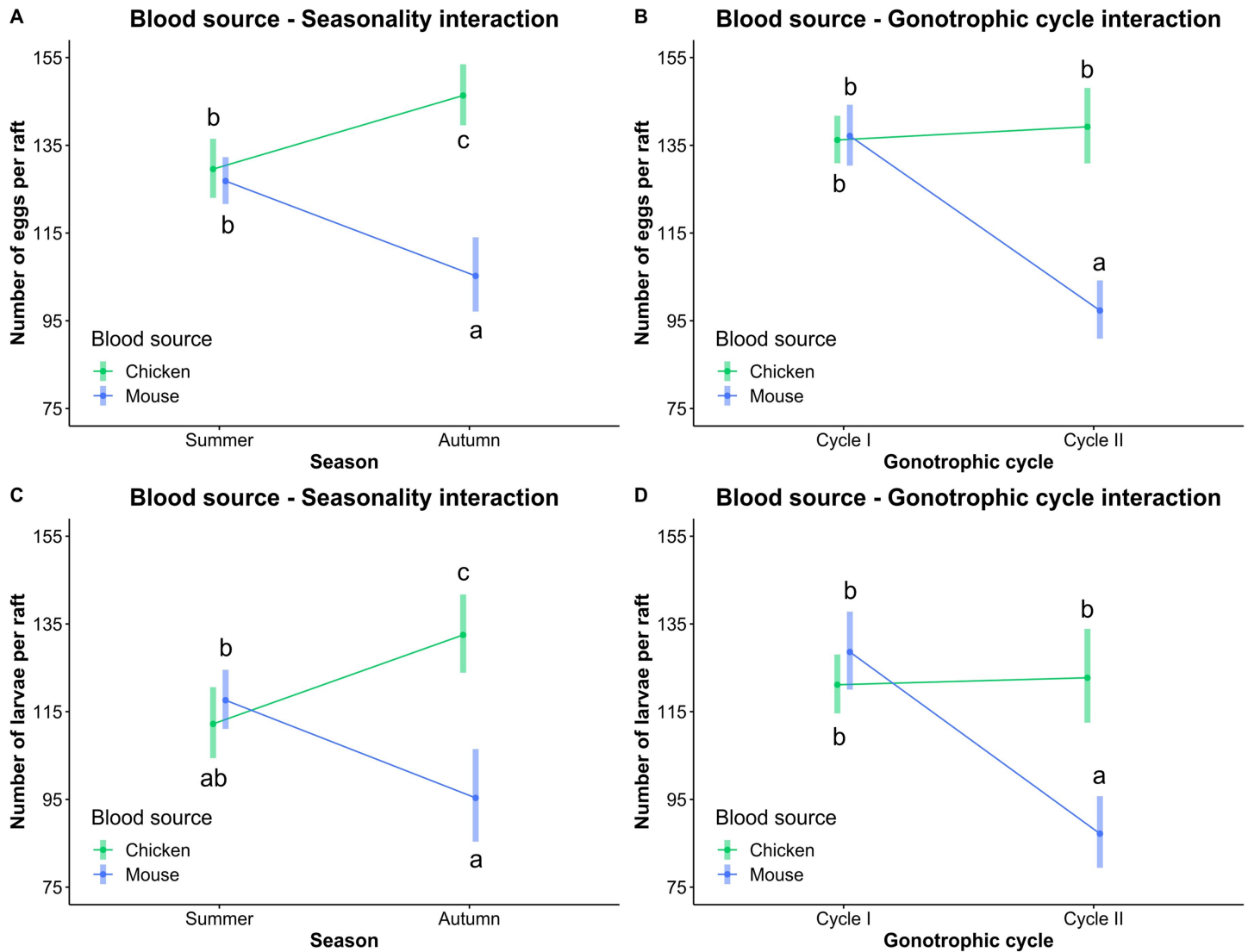


Figure 1.

Interaction plots showing the marginal effects of blood meal source-seasonality and blood meal source-gonotrophic cycle on (A, B) fecundity and (C, D) fertility. Predicted mean values and its corresponding confidence intervals are shown. Means sharing same letters are not statistically different.

Bloodmeal source	Season	Full engorged females	Number of egg rafts	Fecundity (eggs/raft)	Fertility (larvae/raft)	Hatchability (larvae/eggs)
First gonotrophic cycle (first feeding)						
Mouse	Autumn	58	23	124.5 ± 20.9	120.7 ± 21.5	0.96 ± 0.046
Mouse	Summer	208	120	150.7 ± 29.4	141.8 ± 33.9	0.92 ± 0.10
Chicken	Autumn	163	76	147.5 ± 31.8	133.4 ± 38.4	0.89 ± 0.14
Chicken	Summer	116	74	125.6 ± 39.6	109.9 ± 44.0	0.85 ± 0.20
Second gonotrophic cycle (second feeding)						
Mouse	Autumn	23	15	89.1 ± 19.3	73.3 ± 28.0	0.82 ± 0.25
Mouse	Summer	92	45	106.7 ± 32.0	98.8 ± 31.4	0.91 ± 0.10
Chicken	Autumn	73	36	142.1 ± 36.6	129.2 ± 43.7	0.88 ± 0.18
Chicken	Summer	38	27	138.2 ± 30.3	117.3 ± 38.1	0.84 ± 0.18

Table 1.

Results of reproductive outputs of *Culex quinquefasciatus* for the first and second gonotrophic cycle. Mean values are accompanied with its respective standard deviation.

Factor	LRT X ²	df	p-value
Seasonality	15.72	1	7.361×10^{-5}
Blood source	8.12	1	0.00438
Gonotrophic cycle	0.55	1	0.45904
Seasonality:Blood source	25.00	1	7.717×10^{-7}
Seasonality:Gonotrophic cycle	3.11	1	0.07764
Blood source:Gonotrophic cycle	9.29	1	0.00230
Seasonality:Blood source:Gonotrophic cycle	1.42	1	0.23340

Table 2.

Analysis of deviance table of the negative binomial model for the effect of blood meal source, seasonality and gonotrophic cycle on fecundity.

Factor	LRT X ²	df	p-value
Seasonality	11.87	1	0.00057
Blood source	1.64	1	0.19964
Gonotrophic cycle	0.20	1	0.65066
Seasonality:Blood source	13.82	1	0.00020
Seasonality:Gonotrophic cycle	0.87	1	0.34972
Blood source:Gonotrophic cycle	11.40	1	0.00073
Seasonality:Blood source:Gonotrophic cycle	0.04	1	0.83479

Table 3.

Analysis of deviance table of the negative binomial model for the effect of blood meal source, seasonality and gonotrophic cycle on fertility.

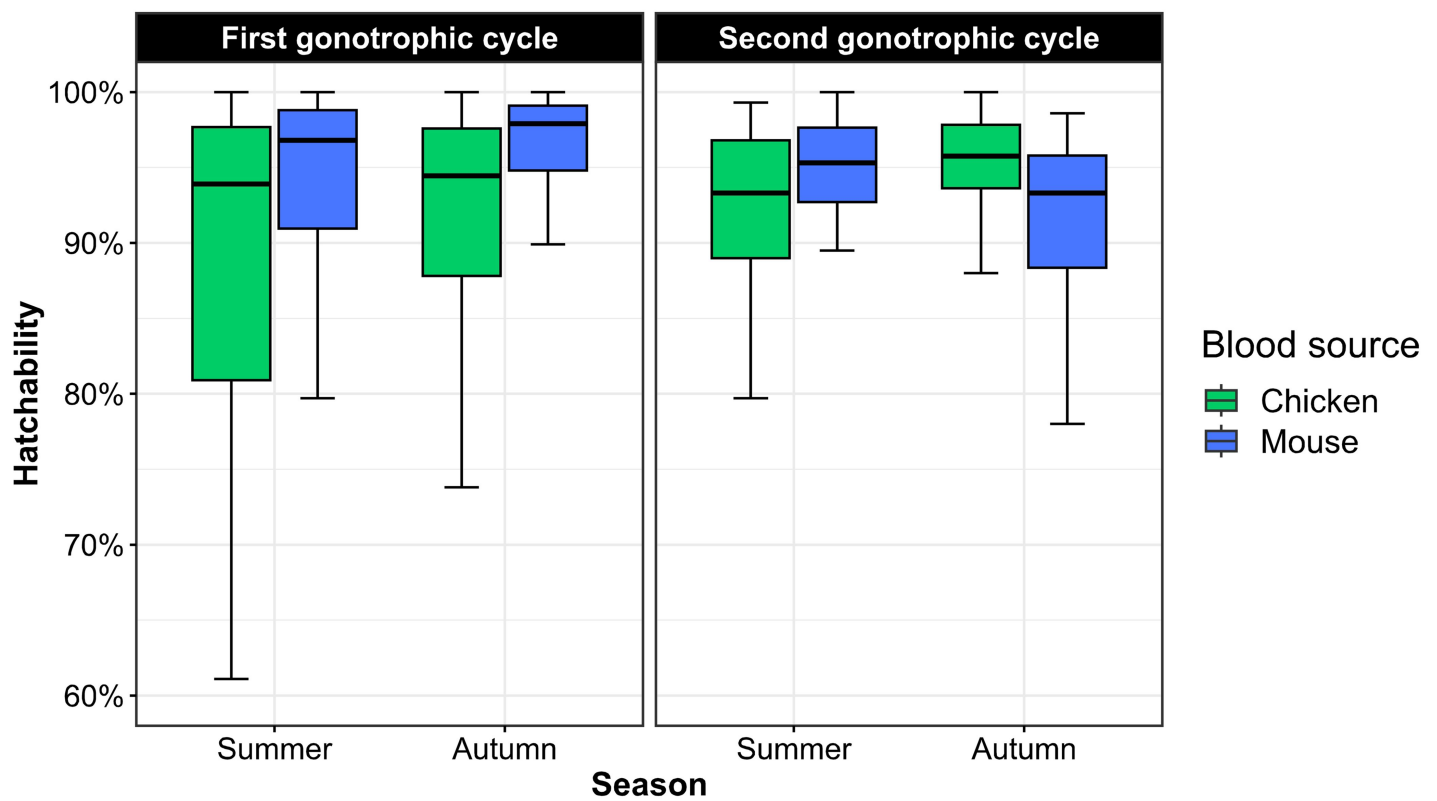


Figure 2.

Boxplots showing the effects of both blood meal source and seasonality on hatchability for the first and second gonotrophic cycle.

are no available studies that have attempted to assess the interaction effect among these factors, and most articles tend to investigate the effects of blood meal source and temperature or photoperiod independently.

Overall, we detected that chicken-fed mosquitoes produced more eggs and larvae per raft than mouse-fed mosquitoes, except in summer where differences were not detected. Akoh et al. (1993) [\[1\]](#) reported a greater fecundity for mosquitoes fed on chicken than those fed on mice. Richards et al. (2012) [\[2\]](#) and Telang & Skinner (2019) found a greater fecundity and fertility in chicken-fed mosquitoes than fed on bovine and pig blood, respectively. This effect is also present in other species of *Culex* (Shroyer & Silvery, 1972 [\[3\]](#)) and *Aedes* (Harrison et al., 2021 [\[4\]](#)). However, other authors have found the opposite pattern with human-fed mosquitoes (Ferdousi & Islam, 2005 [\[5\]](#)) and Guinea pig and human-fed mosquitoes (Xue et al., 2009 [\[6\]](#)) compared with mosquitoes fed upon chickens. In summary, with some few exceptions, mosquitoes fed on avian blood commonly produce a considerably greater batch size and fertility than mammal-fed mosquitoes. We also observed an increase of egg production and hatching per raft from summer to autumn in mosquitoes fed on chicks, whereas we found a decrease of these reproductive traits in mosquitoes fed upon mice. Regarding gonotrophic cycle, the second cycle was less productive in terms of fecundity and fertility for mosquitoes feeding on mice, however, we did not find differences in these reproductive outputs for mosquitoes fed on chicken. For hatchability, it was observed a greater hatching rate in mouse-fed mosquitoes than chicken-fed mosquitoes, however, the effect was not such marked as in fecundity and fertility.

Several hypotheses (-I) nutritional hypothesis, -II) digestive physiology hypothesis and -III) digestive mechanical hypothesis) have been proposed to explain why avian blood produces a greater fecundity and fertility even in mosquitoes with a known mammal-preferred host choice. The first widely accepted general idea (I) suggests that since birds have larger and nucleated erythrocytes, they provide greater amounts of proteins and energy than the anucleated mammalian erythrocytes, inducing a higher fecundity and fertility in mosquitoes feeding from these animals (Alto et al., 2014 [\[7\]](#)). However, it has been shown that mammalian blood contains more total protein, hemoglobin and hematocrit levels than avian blood, suggesting it is nutritional richer. Even though, mosquitoes such as *Cx. quinquefasciatus* experiments a process known as pre-diuresis, excreting large amounts of water (derived from blood serum), allowing it to concentrate and accumulate more proteins. Hence, it could be possible that mosquitoes compensate the low protein quality of bird blood by consuming larger volumes or feeding for longer times (Erram et al., 2022) which might explain the greater fecundity and fertility while feeding on avian hosts. More than the total protein, it has been shown that it is the amino acid balance which determines the egg production, and amino acid content varies between different animals. For *Aedes aegypti*, eight amino acids are required to ensure egg production, namely, arginine, isoleucine, leucine, lysine, phenylalanine, threonine, tryptophan and valine. Consumption of these amino acids induces the expression of yolk protein precursor genes and the resulting synthesis of the protein vitellogenin, which is absorbed by oocytes and allows the development of embryos and posterior egg laying. It has been demonstrated that from all these amino acids, isoleucine is the limiting one, and Greenberg (1951) found that addition of isoleucine to washed sheep erythrocytes increases egg production and oviposition. However, this effect was not observed with the addition of a mixture of the other amino acids. Since mammalian blood is poor in isoleucine compared with avian blood, it was suggested and proven for several species that isoleucine-rich host blood induces a greater production of eggs by mosquitoes (Greenberg, 1951 [\[8\]](#); Harrison et al., 2021 [\[4\]](#)). The digestive physiology hypothesis (II) states that differences in egg production between avian and mammalian hosts is due to variation in digestion rates. Downe et al. (1963) [\[9\]](#) found that in *Mansonia perturbans* digestion of avian blood occurred more rapidly than mammalian blood and this pattern was also confirmed in *Culex tarsalis* (Downe and Archer, 1975 [\[10\]](#)). In *Aedes aegypti*, nutrients from a blood meal can enter the oocyte with a period of 36-48 hs, after that the pinocytotic activity of the gamete decreases and the intake of exterior material is blocked. Because of this, those species of mosquitoes that can achieve the digestion of blood in a short period will

ensure the exploitation of nutrients generating a greater egg production. This could be the situation present in mosquitoes fed upon birds, which produce a greater fecundity (Downe & Archer, 1975 [\[1\]](#)). A third possible explanation is the digestive mechanical hypothesis (III). Mosquitoes need to retain erythrocytes in midgut and hemolyzed the blood meal to exploit proteins. Female mosquitoes are armed with a set of sclerotized teeth-like structures present in the anterior foregut (= cibarial armature) and posterior midgut (= pyloric armature), the first one acts as the primary site of hemolysis, whereas the second armature prevents blood to escape outside the midgut, enhancing the exploitation of resources from red blood cells. Since the structure of cibarial and pyloric armature varies between mosquitoes and because erythrocytes from host are of different shapes and sizes, these factors could contribute to mosquitoes to select those hosts whose blood is easier to hemolyze and its protein easier to assimilate (Lyimo & Ferguson, 2009 [\[2\]](#); Vaughan et al., 1991 [\[3\]](#)). In summary, several factors regarding blood quality and digestive physiology make bird blood a higher nutritional resource for egg production, and despite these factors were usually studied independently, it must be the combination or interaction of all of them which enhance the production of a greater fitness in mosquitoes feeding on birds.

Regarding environmental conditions, there is plentiful information available about the effects of temperature and photoperiod on physiological parameters (survival rate, development time, fecundity, fertility, behavior, etc.) (Hoffman, 1985 [\[4\]](#)). In nature, mosquitoes are exposed to a cyclic variation of both temperature and day length. It can be recognise a warm phase (= thermophase) which coincides with a photophase (light presence) and a cool phase (= cryophase) coinciding with a scotophase (light absence) (Joshi, 1996 [\[5\]](#)). Field and laboratory data have established that all these physiological traits follow a unimodal or hump-shaped relationship in ectothermic organisms, increasing from zero at a thermal minimum to a highest optimum and then decreasing to zero again at a thermal maximum. The best performance of animals generally is obtained at a narrow “optimal range”, which varies among and within species and traits. In *Culex quinquefasciatus* the fecundity curve ranges from a thermal minimum of 5°C to a maximum of 37.7°C with an optimal temperature of 21.4°C (Mordecai et al., 2019 [\[6\]](#)). However, the greatest egg production is restricted to a narrow interval around the thermal optimum. This narrow band has been determined for some species, for *Culex pipiens* complex and *Aedes aegypti* it seems that the optimal temperature ranges from 20°C to 30°C (Spanoudis et al., 2019 [\[7\]](#); Bader et al. 2012).

The effect of temperature on physiological traits has been studied in two ways, keeping it constant across all the experiments or fluctuating it in different ways, generally between a minimum and maximum. For different populations of *Culex quinquefasciatus* and for a constant temperature, Mogi (1992) [\[8\]](#) has found that fecundity increases from 15°C to 20°C and decreases to 28°C. Same pattern was registered for *Aedes albopictus*, with a peak at 26°C and decreasing to 30°C. Fluctuating temperature has the same pattern, although because temperature is allowed to fluctuate within a range mosquitos seem to cope better the extreme conditions, for example, *Aedes albopictus* produce same quantity of eggs at 18°C-26°C and 22°C-30°C (Joshi, 1996 [\[5\]](#)). Regarding photoperiod, fewer studies are available, some of them are those of Mogi (1992) [\[8\]](#) and Costanzo et al. (2015), however, the effects are not as evident as temperature, and while Mogi (1992) [\[8\]](#) found that mid and long day lengths induced greater fecundity, Costanzo et al. (2015) [\[9\]](#) did not find differences of day length on fecundity, then, data are more contradictory than the effects of temperature.

Two important mosquito traits that might affect fecundity and the subsequent fertility are calendar age (= age post-emerging) and physiological age (= gonotrophic cycle). In general, fecundity and fertility decline with calendar age (Akoh et al., 1993 [\[10\]](#); McCann et al., 2009 [\[11\]](#)) and gonotrophic cycle (Richards et al., 2012 [\[12\]](#); Telang & Skinner, 2019 [\[13\]](#)). This general pattern was shown by us in mouse-fed mosquitoes. Although we also recorded a numerical increase in fecundity and fertility between gonotrophic cycles for mosquitoes fed on chicken, this difference was not significant. Same pattern was found by Bennet (1970) in *Aedes aegypti* fed upon avian and mammalian hosts. In addition, Obholz (unpublished data) also found an increase in fecundity and

fertility for *Ae. Aegypti* fed on mice. A difference in egg production may be attributable to variation in feeding rate, but since we only evaluated full-engorged female mosquitoes this difference cannot be explained based on this rate. Hence, there must be an underlying physiological or environmental process that may be affecting fecundity and fertility and further studies are needed to corroborate it.

All studies assessing the effect of temperature or photoperiod on reproductive traits have been independent from the source of blood meal, however, our results suggest that these two environmental variables have a significant effect on fecundity and fertility, affecting differentially regarding blood meal source. We found that in summer (mean temperature = 25°C and long-day photoperiod) there is not differences on fecundity and fertility for mosquitoes fed upon mouse and chicken, however, in autumn (mean temperature = 19°C and short-day photoperiod) the fecundity and fertility of chicken-fed mosquitoes increased whereas in mouse-fed mosquitoes decreased. These results are not directly comparable with literature, since there are no studies that have evaluated the interaction effect of blood meal source and seasonality (temperature + photoperiod). Several authors suggest that fecundity and fertility increase at higher temperatures and photoperiod for both bird and mammal-fed mosquitoes (Abdallah et al., 2020; Al-Rashidi et al., 2022 [↗](#), Shehata, 2018 [↗](#)), however, a difference based on blood meal source is elusive. This little evidence makes it evident that more studies are needed to assess the combined effect of temperature, photoperiod and blood source on reproductive traits since a common pattern has not been readily identified at the moment.

Fertility in our experiments varied among treatments. Considering the number of eggs, differences in fertility may be attributable to two factors: (I) variation in fecundity or (II) variation in hatching rate (= hatchability). The first has already been tested, then, to assess if the variation in number of larvae was a result of differences in hatchability, we evaluated if there were changes in the percentage of eggs hatched. We find significant differences in hatchability regarding the type of blood meal source, obtaining a mean hatching rate for mammal-fed mosquitoes of 92% and 87% for bird-feeding mosquitoes. In literature, hatchability (also cited as viability or even fertility) has been shown to vary according to different temperatures (van der Linde et al., 1990 [↗](#)) but not with blood meal sources (dos Santos Dias et al., 2018; Richard et al., 2012). Our preliminary data suggest that changes in fertility are a consequence of variation in egg production due to the effects of blood meal source and seasonality and not in hatchability.

Understanding factors leading host selection and host switch are critical to predict arboviral outbreaks and arbovirus spill over. It is well known that many species of *Culex* mosquitoes experience changes in their host usage, from birds in spring and late summer to mammals in late summer and autumn, which usually trigger the increase of viral infection reported on humans, turning from a period of viral amplification to viral transmission. Nevertheless, studies considering the causes of host switching are very scarce and they mainly focus on host factors, such as host availability or defensive behavior (Kilpatrick et al., 2006 [↗](#), Burkett-Cadena et al., 2011 [↗](#)). A variety of mosquito physiological traits have been studied as an attempt to explain host selection but neither of them have been applied in host switch mechanisms (Lyimo & Ferguson, 2009 [↗](#); Yan et al., 2017 [↗](#), 2018 [↗](#)). In this work we investigated for the first time if some reproductive traits could explain the pattern in host shift from birds to mammals. In our study we included the interaction effects of blood meal source and seasonality on fecundity and fertility, two variables analyzed separately but never combined, to the best of our knowledge. Our results suggest that there is a change in fecundity and fertility through seasons due to the effects of blood meal source and seasonality.

In summary, our model suggests that biological and climatic factors induce physiological changes in mosquitoes which triggers selection of new hosts, generating an increase of fitness (fertility and fecundity). Our results did not support the hypothesis that greater fitness on birds is present in

summer and on mammals in autumn, since we register the opposite pattern. Notwithstanding, since *Culex* mosquitoes have a wide range of host niche it remains inconclusive whether other factors may be intervening in mosquitoes seasonal host shift.

Acknowledgements

We would like to thank Dr. Agustín Quaglia for providing valuable help in statistical analysis. In addition, we greatly appreciate very helpful comments of the manuscript by anonymous reviewers. This study was supported by grant PICT 2018-1172 (FONCYT) and Consolidar Secyt-UNC (2018-2023).

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Author information

Kevin Alen Rucci

Laboratorio de Arbovirus, Instituto de Virología “Dr. J. M. Vanella” (InViV), Facultad de Ciencias Médicas (FCM), Universidad Nacional de Córdoba (UNC), Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina

For correspondence: kevin.rucci@mi.unc.edu.ar

ORCID iD: [0000-0002-0940-2423](https://orcid.org/0000-0002-0940-2423)

Mariana Pueta

Instituto de Investigaciones en Biodiversidad y Medioambiente (INIBIOMA), Universidad Nacional de Comahue (UnCo), Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina

ORCID iD: [0000-0002-9837-4718](https://orcid.org/0000-0002-9837-4718)

Adrián Díaz

Laboratorio de Arbovirus, Instituto de Virología “Dr. J. M. Vanella” (InViV), Facultad de Ciencias Médicas (FCM), Universidad Nacional de Córdoba (UNC), Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina

For correspondence: adrian.diaz@fcm.unc.edu.ar

ORCID iD: [0000-0001-5953-2907](https://orcid.org/0000-0001-5953-2907)

Editors

Reviewing Editor

Chima Nwaogu

University of Cape Town, South Africa

Reviewer #1 (Public Review):

Summary:

This study examines the role of host blood meal source, temperature, and photoperiod on the reproductive traits of *Cx. quinquefasciatus*, an important vector of numerous pathogens of medical importance. The host use pattern of *Cx. quinquefasciatus* is interesting in that it feeds on birds during spring and shifts to feeding on mammals towards fall. Various hypotheses have been proposed to explain the seasonal shift in host use in this species but have provided limited evidence. This study examines whether the shifting of host classes from birds to mammals towards autumn offers any reproductive advantages to *Cx. quinquefasciatus* in terms of enhanced fecundity, fertility, and hatchability of the offspring. The authors found no evidence of this, suggesting that alternate mechanisms may drive the seasonal shift in host use in *Cx. quinquefasciatus*.

Strengths:

Host blood meal source, temperature, and photoperiod were all examined together.

Weaknesses:

The study was conducted in laboratory conditions with a local population of *Cx. quinquefasciatus* from Argentina. I'm not sure if there is any evidence for a seasonal shift in the host use pattern in *Cx. quinquefasciatus* populations from the southern latitudes.

Reviewer #2 (Public Review):

Summary:

Conceptually, this study is interesting and is the first attempt to account for the potentially interactive effects of seasonality and blood source on mosquito fitness, which the authors frame as a possible explanation for previously observed host-switching of *Culex quinquefasciatus* from birds to mammals in the fall. The authors hypothesize that if changes in fitness by blood source change between seasons, higher fitness in birds in the summer and on mammals in the autumn could drive observed host switching. To test this, the authors fed individuals from a colony of *Cx. quinquefasciatus* on chickens (bird model) and mice (mammal model) and subjected each of these two groups to two different environmental conditions reflecting the high and low temperatures and photoperiod experienced in summer and autumn in Córdoba, Argentina (aka seasonality). They measured fecundity, fertility, and hatchability over two gonotrophic cycles. The authors then used a generalized linear mixed model to evaluate the impact of host species, seasonality, and gonotrophic cycle on fecundity and fertility and a null model analysis via data randomization for hatchability. The authors were trying to test their hypothesis by determining whether there was an interactive effect of season and host species on mosquito fitness. This is an interesting hypothesis; if it had been supported, it would provide support for a new mechanism driving host switching. While the authors did report an interactive impact of seasonality and host species, the directionality of the effect was the opposite of that hypothesized. While this finding is interesting and worth reporting, there are significant issues with the experimental design and the conclusions that are drawn from the results, which are described below. These issues should be addressed to make the findings trustworthy.

Strengths:

1. Using a combination of laboratory feedings and incubators to simulate seasonal environmental conditions is a good, controlled way to assess the potentially interactive impact of host species and seasonality on the fitness of *Culex quinquefasciatus* in the lab.
2. The driving hypothesis is an interesting and creative way to think about a potential driver of host switching observed in the field.

Weaknesses:

1. There is no replication built into this study. Egg lay is a highly variable trait, even within treatments, so it is important to see replication of the effects of treatment across multiple discrete replicates. It is standard practice to replicate mosquito fitness experiments for this reason. Furthermore, the sample size was particularly small for some groups (e.g. 15 egg rafts for the second gonotrophic cycle of mice in the autumn, which was the only group for which a decrease in fecundity and fertility was detected between 1st and 2nd gonotrophic cycles). Replicates also allow investigators to change around other variables that might impact the results for unknown reasons; for example, the incubators used for fall/summer conditions can be swapped, ensuring that the observed effects are not artifacts of other differences between treatments. While most groups had robust sample sizes, I do not trust the replicability of the results without experimental replication within the study.
2. Considering the hypothesis is driven by the host switching observed in the field, this phenomenon is discussed very little. I do not believe *Cx. quinquefasciatus* host switching has been observed in Argentina, only in the northern hemisphere, so it is possible that the species could have an entirely different ecology in Argentina. It would have been helpful to conduct a blood meal analysis prior to this experiment to determine whether using an Argentinian population was appropriate to assess this question. If the Argentinian populations don't experience host switching, then an Argentinian colony would not be the appropriate colony to use to assess this question. Given that this experiment has already been conducted with this population, this possibility should at least be acknowledged in the discussion. Or if a study showing host switching in Argentina has been conducted, it would be helpful to highlight this in the introduction and discussion.
3. The impacts of certain experimental design decisions are not acknowledged in the manuscript and warrant discussion. For example, the larvae were reared under the same conditions to ensure adults of similar sizes and development timing, but this also prevents mechanisms of action that could occur as a result of seasonality experienced by mothers, eggs, and larvae.
4. There are aspects of the data analysis that are not fully explained and should be further clarified. For example, there is no explanation of how the levels of categorical variables were compared.
5. The results show the opposite trend as was predicted by the authors based on observed feeding switches from birds to mammals in the autumn. However, they only state this once at the end of the discussion and never address why they might have observed the opposite trend as was hypothesized.
6. Generally speaking, the discussion has information that isn't directly related to the results and/or is too detailed in certain parts. Meanwhile, it doesn't dig into the meaning of the results or the ways in which the experimental design could have influenced results.
7. Beyond the issue of lack of replication limiting trust in the conclusions in general, there is one conclusion reached at the end of the discussion that would not be supported, even if additional replicates are conducted. The results do not show that physiological changes in mosquitoes trigger the selection of new hosts. Host selection is never measured, so this claim cannot be made. The results don't even suggest that fitness might trigger selection because the results show that physiological changes are in the opposite direction as what would be hypothesized to produce observed host switches. Similarly, the last sentence of the abstract is

not supported by the results.

8. Throughout the manuscript, there are grammatical errors that make it difficult to understand certain sentences, especially for the results.

This study is driven by an interesting question and has the potential to be a valuable contribution to the literature.

Author Response

Reviewer #1 (Public Review):

The study was conducted in laboratory conditions with a local population of Cx. quinquefasciatus from Argentina. I'm not sure if there is any evidence for a seasonal shift in the host use pattern in Cx. quinquefasciatus populations from the southern latitudes.

Unfortunately, studies conducted in South America to understand host use by Culex mosquitoes are very limited, and there are virtually no studies on the seasonal pattern of host use. In Argentina, there is some evidence (Stein et al., 2013; Beranek, 2018) regarding the seasonal change in host use by Culex species, including Culex quinquefasciatus, where the inclusion of mammals during the autumn has been observed. As part of a comprehensive study on characterizing bridge vectors for SLE and WN viruses, our research group is currently working on the molecular identification of blood meals from engorged females to gain deeper insights into the seasonal host use by Culex mosquitoes.

While the seasonal change in host use by Culex quinquefasciatus has not been reported in Argentina so far, there has been an observed increase in reported cases of SLE virus in humans between summer and autumn (Spinsanti et al., 2008). It is based on this evidence that we hypothesize there is a seasonal change in host use by Culex quinquefasciatus, similar to what occurs in the United States. This is also considering that both countries (Argentina and the United States) have regions with similar climatic conditions (temperate climates with thermal and hydrological seasonality).

I think the authors need to discuss more about the bigger question they were addressing. I think that the discussion section can be strengthened greatly by elaborating on whether there is evidence for a seasonal shift in host use pattern in Cx. quinquefasciatus in the southern latitudes. If yes, what alternate mechanisms they believe could be driving the seasonal change in host use in this species in the southern latitudes now that they show the 'deriving reproductive advantages' hypothesis to be not true for those populations.

We will restructure our discussion to align it with our results, as suggested.

Grammar and writing

The manuscript will be grammatically revised.

Reviewer #2 (Public Review):

There is no replication built into this study. Egg lay is a highly variable trait, even within treatments, so it is important to see replication of the effects of treatment across multiple discrete replicates. It is standard practice to replicate mosquito fitness experiments for this reason. Furthermore, the sample size was particularly small for some groups (e.g. 15

egg rafts for the second gonotrophic cycle of mice in the autumn, which was the only group for which a decrease in fecundity and fertility was detected between 1st and 2nd gonotrophic cycles). Replicates also allow investigators to change around other variables that might impact the results for unknown reasons; for example, the incubators used for fall/summer conditions can be swapped, ensuring that the observed effects are not artifacts of other differences between treatments. While most groups had robust sample sizes, I do not trust the replicability of the results without experimental replication within the study.

We agree egg lay is a variable trait and so we consider high numbers of mosquitoes and egg lay during experiments compared to our studies of the same topics. Evaluating variables such as fecundity, fertility, or other types of variables (collectively referred to as "life tables") is a challenging issue that depends on several intrinsic and extrinsic factors. Because of all of this, in some experiments, sample sizes might not be very large, and in several articles, lower sample sizes could be found. For instance, in Richards et al. (2012), for *Culex quinquefasciatus*, during the second gonotrophic cycle, some experiments had 13 or even 6 egg rafts. For species like *Aedes aegypti*, the sample size for life table analysis is also usually small. As an example, Muttis et al. (2018) reported between 1 and 4 engorged females (without replicates). Because of this, we do find our sample sizes quite robust for our results.

Regarding the need to repeat the experiments in order to give more robustness to the study we also agree. However, after a review of the literature (articles cited in the original manuscript), it is apparent that similar experiments are not frequently repeated as such. Examples of this are the studies of Richards et al. (2012), Demirci et al. (2014) or Telang & Skinner (2019), which even manipulate several cages at a time as "replicates", they are not true replicates because they summarise and manipulate all data together, and do not repeat the experiment several times. We see these "replicates" as a way of getting a greater N.

As it was stated by the reviewer, repetition is a resource and time consuming activity that we are not able to do. Replicating the experiment poses a significant time challenge. The original experiment took over three months to complete, and it is anticipated that a similar timeframe would be necessary for each replication (6 months in total considering two more replicates). Given our existing commitments and obligations, dedicating such an extensive period solely to this would impede progress on other crucial projects and responsibilities. Given the limitations of resources and time and the infrequent use of experimental repetition in this type of studies, we suggest performing a simulation-based analysis. This approach involves generating synthetic data that mimics the expected characteristics of the original experiment and subsequently subjecting it to the same analysis routine. The main goal of this simulation will be to evaluate the potential spuriousness and randomness of the results that might arise due to the experimental conditions. We will introduce this simulation-based analysis in the next revised version of the manuscript.

*Considering the hypothesis is driven by the host switching observed in the field, this phenomenon is discussed very little. I do not believe *Cx. quinquefasciatus* host switching has been observed in Argentina, only in the northern hemisphere, so it is possible that the species could have an entirely different ecology in Argentina. It would have been helpful to conduct a blood meal analysis prior to this experiment to determine whether using an Argentinian population was appropriate to assess this question. If the Argentinian populations don't experience host switching, then an Argentinian colony would not be the appropriate colony to use to assess this question. Given that this experiment has already been conducted with this population, this possibility should at least be acknowledged in the discussion. Or if a study showing host switching in Argentina has been conducted, it would be helpful to highlight this in the introduction and discussion.*

We are aware that few studies regarding host shifting in South America are available, some such those conducted by Stein et al. (2013) and Beranek (2018) reported a moderate host switch for *Culex quinquefasciatus* in Argentina. We have already performed a study about seasonal host feeding patterns for this species. As you suggested, we could mention it in the discussion to highlight our partial findings. However, even though there are few studies regarding host shifting, our hypothesis is based mainly in the seasonality of human cases of WNV and SLEV, a pattern that has been demonstrated for our region, see for example the study of Spinsanti et al. (2008).

The impacts of certain experimental design decisions are not acknowledged in the manuscript and warrant discussion. For example, the larvae were reared under the same conditions to ensure adults of similar sizes and development timing, but this also prevents mechanisms of action that could occur as a result of seasonality experienced by mothers, eggs, and larvae.

We understand the confusion that may have arisen due to a lack of further details in the methodology. If we are not mistaken, you are referring to our oversight regarding the consideration of carry-over effects of larvae rearing that could potentially impact reproductive traits. When investigating the effects of temperature or other environmental factors on reproductive traits, it is possible to acclimate either larvae or adults. This is due to the significant phenotypic plasticity that mosquitoes exhibit throughout their entire ontogenetic cycle. In our study, we followed an approach similar to that of other authors where the adults are exposed to experimental conditions (temperature and photoperiod). For a similar approach you can refer to the studies conducted by Ferguson et al. (2018) for *Cx. pipiens*, Garcia Garcia & Londoño Benavides (2007) for *Cx. quinquefasciatus* and Christiansen-Jucht et al. (2014, 2015) for *Anopheles gambiae*.

Beyond the issue of lack of replication limiting trust in the conclusions in general, there is one conclusion reached at the end of the discussion that would not be supported, even if additional replicates are conducted. The results do not show that physiological changes in mosquitoes trigger the selection of new hosts. Host selection is never measured, so this claim cannot be made. The results don't even suggest that fitness might trigger selection because the results show that physiological changes are in the opposite direction as what would be hypothesized to produce observed host switches. Similarly, the last sentence of the abstract is not supported by the results.

We agree with this observation. However, we did not evaluate the impact of fitness on host selection in this study. Instead, we aimed to investigate the potential influence of seasonality on mosquito fitness as a potential trigger for a shift in host selection. We agree that we have incorrectly used the term “host selection” when we should actually be discussing “host use change”. Our results indicate a seasonal alteration in mosquito fitness in response to temperature and photoperiod changes. Building upon this observation, we will discuss into our hypotheses and theoretical model to explain this seasonal shift in host use.

Grammar and writing

The manuscript will be grammatically revised by a professional translator.